Mrugank Dake

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New York, NY

WORK EXPERIENCE

Postdoctoral Research Scientist, Dartmouth College **June 2026 - present**

Product Data Science Intern, Meta Reality Labs **May 2025 - August 2025**

- Investigated hand usage patterns across the Quest ecosystem using Presto and Spark, querying massive datasets to identify friction points and drive core product insights.
- Developed Machine Learning models in Python to isolate demographic and behavioral factors contributing to user drop-off, leading to targeted UX recommendations for Quest applications.
- Applied Graph Theory approaches to quantify the influence of social networks on Mixed Reality (MR) adoption and long-term hand-tracking retention.
- Collaborated with cross-functional (XFN) partners to refine topline metrics, leveraging data-driven findings to influence the product roadmap for next-generation hardware.
- Actively tested and provided feedback on state-of-the-art VR/MR prototypes, ensuring internal quality standards and alignment with user-centric data trends.

Graduate Teaching Assistant, New York University **January 2023 - December 2025**
Courses: Introduction to Scientic Programming (Graduate), EEG/MEG/iEEG Methods (Graduate), Introduction to Cognitive Neuroscience (Undergraduate), Math Tools 3: Linear Systems & Fourier Transforms (Graduate)

- Led discussions and tutorials for 150+ students, translating complex topics in linear algebra, signal processing, and Fourier transforms into digestible insights.
- Engineered coding assignments and interactive tutorials in Python and MATLAB, focusing on real-world applications of scientific programming and data analysis.
- Developed custom animations to visually demonstrate abstract mathematical concepts, significantly improving student comprehension of linear systems and multidimensional data.
- Managed end-to-end grading workflows for technical assignments and research papers, ensuring rigorous academic standards and providing constructive feedback on experimental design.

EDUCATION

New York University New York, NY
PhD in Computational Cognitive Neuroscience **May 2026**
MPhil in Experimental Psychology **May 2025**
GPA: 3.95/4.0
Relevant Coursework: Math Tools for Cognitive & Neural Science, Psychophysics, Data Science, Computational Neuroscience

Indian Institute of Science Education & Research (IISER) Tirupati, India
BS-MS Dual Degree (Physics & Biology) **August 2021**
CGPA: 8.2/10.0
Relevant Coursework: Nonlinear Dynamics & Chaos, Linear Algebra (I & II), Calculus (Univariate and Multivariate), Statistical Thermodynamics, Data Science

RESEARCH EXPERIENCE

Clayspace Lab, New York University **September 2021 - Present**
Doctoral Researcher, Supervisor: Dr. Clayton Curtis

- High-Performance Computing: Optimized population receptive field (pRF) mapping for fMRI data by implementing parallel processing across CPUs and GPUs (CUDA) to resolve computational bottlenecks.
- Multimodal Data Fusion: Led end-to-end pipelines for high-dimensional datasets including human behavior, eye-tracking, and neural signals (fMRI, MEG, ECoG, EEG).
- Neural Simulations: Developed complex electric field simulations (SimNIBS) and utilized TMS to model distributed neural representations and their causal role in working memory.
- Academic Impact: Lead author of a manuscript published in *Nature Communications*; currently spearheading five additional projects targeting top-tier journals and conferences; active peer-reviewer for CogSci, CCN, ICLR and eLife
- Scientific Communication: Presented research at Stanford, Dartmouth, and international conferences (SfN, CCN) to translate complex neural modeling for both technical and non-technical audiences.

Sciurid Lab, IISER Tirupati **May 2020 - August 2021**
Masters Thesis, Supervisor: Dr. Nandani Rajamani

- Acoustic Deep Learning: Developed and evaluated CNNs and template-matching classification models to automate the detection of rare species vocalizations in noisy rainforest environments.
- Citizen Science & Big Data: Managed a national-scale project involving the ingestion, annotation, and analysis of large-scale acoustic datasets using Python and R.
- Statistical Modeling: Conducted rigorous hypothesis testing and feature extraction on passive acoustic data, resulting in a completed thesis and two manuscripts in preparation.

iGEM Team Lead, IISER Tirupati **November 2017 – November 2019**

- Technical Leadership: Directed a multidisciplinary team of 17 researchers to engineer "*CoCa coli*," a probiotic immunotherapy for targeted colon cancer detection, earning a Gold Medal at the Giant Jamboree in Boston (top-tier of 300+ teams).
- Project Execution: Secured INR 1million ($12,000 USD) in research funding via the Indian Biological Engineering Competition (iBEC) to support end-to-end genetic circuit design and in-vitro validation.
- Operational Management: Managed international logistics and cross-functional coordination, applying first-principles biological engineering to deliver a validated scientific product on a global stage.

VOLUNTEER WORK

Website Coordinator, ClimateMatch Academy **February 2023 - July 2023**

- Cross-Institutional Collaboration: Engineered and maintained the official website for the ClimateMatch curriculum in partnership with Neuromatch Academy, streamlining content delivery for global participants.

Cognition & Perception Open House Student Organizer, New York University **September 2022 - June 2023**

- Strategic Event Leadership: Elected by the student body to spearhead recruitment events for the incoming 2023 PhD cohort, managing logistics for 40+ international candidates.
- Operational Logistics: Orchestrated end-to-end event planning, including travel, accommodation, and interview scheduling, ensuring a seamless experience for faculty and prospective students.

AWARDS

Society for Neuroscience Trainee Professional Development Award ($1,000)	**2025**
Dean's Award for Research Communication Excellence, NYU ($2,000)	**2025**
MacCracken Doctoral Fellowship, NYU	**2021 – 2026**
Indian Biological Engineering Competition (iBEC) Grant (~$12,000)	**2020**
Indian Biological Engineering Competition (iBEC) Grant (~$12,000)	**2019**
GSAS Conference Travel Grant, NYU	**2025**
INSPIRE Fellowship, Department of Science and Technology India	**2016 – 2021**
5th Place, Indian National Brain Bee (INBB)	**2015**

SKILLS

Programming
Python, MATLAB, R, SQL, LaTeX, CSS, JavaScript, HTML, React, NodeJS, Supabase, Vercel, Antigravity, VSCode

Data Science & ML
PyTorch, TensorFlow, Keras, Scikit-learn, Neural Networks, Time-series Analysis

Neuroscience Tools
EEG, MEG, fMRI, TMS, ECoG, Fieldtrip, MNE-Python, SimNIBS, Psychtoolbox

Other Tools & Languages
Git/GitHub, Adobe Illustrator, Adobe Photoshop
English (Fluent), Hindi, Marathi